

COMPUTER SCIENCE TRIPOS Part IB – 2024 – Paper 6

8 Logic and Proof (mj201)

Resolution

- (a) Consider the following formulae, where a, b and c are constants and v, w, x, y and z are variables:

$$P(a, v) \rightarrow \neg Q(b, w) \quad (1)$$

$$\neg(\neg Q(b, x) \wedge P(b, y)) \quad (2)$$

$$\neg(\neg P(z, z) \wedge \neg P(z, c)) \quad (3)$$

- (i) Convert the formulae above into conjunctive normal form (CNF) and express the result as a set of clauses. State which rule you used for each conversion step. [3 marks]

Answer:

$$\begin{aligned} P(a, v) \rightarrow \neg Q(b, w) \\ \Downarrow X \rightarrow Y \Rightarrow \neg X \vee Y \\ \neg P(a, v) \vee \neg Q(b, w) \end{aligned}$$

$$\begin{aligned} \neg(\neg Q(b, x) \wedge P(b, y)) \\ \Downarrow \neg(X \wedge Y) \Rightarrow \neg X \vee \neg Y \\ \neg\neg Q(b, x) \vee \neg P(b, y) \\ \Downarrow \neg\neg X \Rightarrow X \\ Q(b, x) \vee \neg P(b, y) \end{aligned}$$

$$\begin{aligned} \neg(\neg P(z, z) \wedge \neg P(z, c)) \\ \Downarrow \neg(X \wedge Y) \Rightarrow \neg X \vee \neg Y \\ \neg\neg P(z, z) \vee \neg\neg P(z, c) \\ \Downarrow \neg\neg X \Rightarrow X \\ P(z, z) \vee P(z, c) \end{aligned}$$

Clauses: $\{\neg P(a, v) \vee \neg Q(b, w), Q(b, x) \vee \neg P(b, y), P(z, z) \vee P(z, c)\}$ or can also be expressed as $\{\{\neg P(a, v), \neg Q(b, w)\}, \{Q(b, x), \neg P(b, y)\}, \{P(z, z), P(z, c)\}\}$.

- (ii) Convert the clauses resulting from part (a)(i) into Kowalski form. [2 marks]

Answer: In Kowalski form the antecedent of an implication is a conjunction of negative literals in a clause, and the consequent is a disjunction of positive literals. (Definition not required for marks.)

$$P(a, v) \wedge Q(b, w) \rightarrow \quad (1)$$

$$P(b, y) \rightarrow Q(b, x) \quad (2)$$

$$\rightarrow P(z, z) \vee P(z, c) \quad (3)$$

- (iii) Using the clauses resulting from part (a)(ii), give a resolution proof for them. Use clause (1) as the top clause. Indicate the selected literal(s), clause and substitution used at each step. [7 marks]

Answer: This is one possible proof:

$$\begin{array}{llll}
 (3) & P(a, v) \wedge \underline{Q(b, y)} & \rightarrow & \\
 & \downarrow & (2) & \{\} \\
 (4) & P(a, v) \wedge \underline{P(b, y)} & \rightarrow & \\
 & \downarrow & (3) & \{b/z, c/y\} \\
 (5) & P(a, v) & \rightarrow & \underline{P(b, b)} \\
 & \downarrow & (4) & \{b/y\} \text{ ancestral step} \\
 (6) & \underline{P(a, v')} \wedge \underline{P(a, v'')} & \rightarrow & \\
 & \downarrow & (3) & \{a/z, a/v', c/v''\} \text{ full resolution} \\
 & \rightarrow & &
 \end{array}$$

Decision
procedures

- (b) (i) Convert the following formulae into clauses:

$$M \rightarrow (N \rightarrow M) \quad (4)$$

$$M \rightarrow (N \vee P) \quad (5)$$

$$N \rightarrow (\neg Q \wedge \neg R) \quad (6)$$

$$P \rightarrow (R \wedge \neg Q) \quad (7)$$

$$M \quad (8)$$

[2 marks]

Answer:

$$M \rightarrow (N \rightarrow M) \Rightarrow \neg M \vee \neg N \vee M$$

$$M \rightarrow (N \vee P) \Rightarrow \neg M \vee N \vee P$$

$$N \rightarrow (\neg Q \wedge \neg R) \Rightarrow \neg N \vee (\neg Q \wedge \neg R) \Rightarrow (\neg N \vee \neg Q) \wedge (\neg N \vee \neg R)$$

$$P \rightarrow (R \wedge \neg Q) \Rightarrow \neg P \vee (R \wedge \neg Q) \Rightarrow (\neg P \vee R) \wedge (\neg P \vee \neg Q)$$

$$M$$

Clauses: $\{\{\neg M, \neg N, M\}, \{\neg M, N, P\}, \{\neg N, \neg Q\}, \{\neg N, \neg R\}, \{\neg P, R\}, \{\neg P, \neg Q\}, \{M\}\}$.

- (ii) Use the DPLL method to find a model satisfying the clauses from part (b)(i), or to prove that no such model exists. Briefly explain your work in each step. [6 marks]

Answer: There are a number of ways to answer this, but DPLL is clearly invited to be used, and other ad hoc methods will not get full marks.

$$\{\neg M, \neg N, M\} \quad \{\neg M, N, P\} \quad \{\neg N, \neg Q\} \quad \{\neg N, \neg R\} \quad \{\neg P, R\} \quad \{\neg P, \neg Q\} \quad \{M\}$$

Tautological elimination for M and $\neg M$:

$$\{\neg M, N, P\} \quad \{\neg N, \neg Q\} \quad \{\neg N, \neg R\} \quad \{\neg P, R\} \quad \{\neg P, \neg Q\} \quad \{M\}$$

— *Solution notes* —

Unit M , so M is true:

$$\{N, P\} \quad \{\neg N, \neg Q\} \quad \{\neg N, \neg R\} \quad \{\neg P, R\} \quad \{\neg P, \neg Q\}$$

Pure $\neg Q$, so Q is false:

$$\{N, P\} \quad \{\neg N, \neg R\} \quad \{\neg P, R\}$$

There are no unit clauses and no pure literals, so a case split is forced. For example, we can try if N is true (unit):

$$\{\neg R\} \quad \{\neg P, R\}$$

Unit $\neg R$, so R is false:

$$\{\neg P\}$$

Now unit $\neg P$ (also pure) so P is false:

we have all clauses deleted.

Case split for N is false is not needed.

The original set of clause is satisfiable as no clauses are left (all deleted). No need to do the second branch of the case split. Other runs of the methods are possible. The model is: $M = \text{true}$, $N = \text{true}$, $P = \text{false}$, $Q = \text{false}$, $R = \text{false}$.
